speed, variable-flow primary pumping to enhance system efficiency. Depending on locale, system size, and selection criteria for the chiller, compound pumping may still be beneficial to system design and operating stability. Chiller selections often are limited to maximum fluid velocities of 11 fps, and minimum velocities of 1.5 to 2 fps. Variable-speed pumping in both primary and secondary circuits can be applied to reduce mixing that might occur as chillers are sequenced on and off to compensate for the loads.

The following are some advantages of compound circuits:

- They allow different water temperatures and temperature ranges in different elements of the system. Compound pumping can be used to vary coil capacity by controlling leaving coil air temperature, which leads to better latent energy control.
- They decouple the circuits hydraulically, thereby simplifying the control, operation, and analysis of large systems. Hydraulic decoupling also prevents unwanted series or parallel operation of pumps. Large water circuits often experience pressure imbalances, but compound pumping allows a design to be hydraulically organized into separate smaller subsystems, making troubleshooting and operation easier.

Variable-Speed Pumping Application

Centrifugal pumps may also be operated with variable-frequency drives, which adjust the speed of the electric motor, changing the pump curve. In this application, a pump controller and typically one frequency drive per pump are applied to control the required system variable.

The most typical application is to control differential pressure across one or more branches of the piping network. In a common application, the differential pressure is sensed across a control valve or, alternatively, the valve, coil, and branch (Figure 11). The controller is given a set point equal to the pressure loss of the sensed components at design flow. For example, a 5 psi differential pressure was used to size the control valve; the sensor is connected to sense differential pressure across the valve, so a 5 psi set point is given to the controller. As the control valve closes in reaction to a control signal (Figure 12) from 1 to 2, the differential pressure across the branch rises as the system curve shifts counterclockwise up the pump curve. The pump controller, sensing an increase in differential, decreases the speed of the pump to about point 3, roughly 88% speed. Note that each system curve is shown with two representations. The system curve is the simple relationship of the flow ratio squared to the head ratio. Under control of a pump controller with a single sensor across the control valve, as shown in Figure 11, the pump decreases in speed until the theoretical zero-flow point, at which the pump goes just fast enough to maintain a differential pressure under no flow. In this case, the speed is about 88%, and the power is about 68%. In operation, expect there to be a difference in performance. Figure 12 shows the change as a step function, opposite of the way in which the pump controller functions. Based on the control method, the controller adjustment settings (gain, integral time, and derivative time), system hydraulics, valve time constant, sensor sensitivity, etc., it is far more likely that a small incremental rise in differential pressure will have a corresponding small decrease in pump speed, as the valve repositions itself in reaction to its control signal. This appears as a small sawtoothed series of system curve and pump curve intersection points, and is not of real importance. What is important is that many factors influence operation, and theoretical variable-speed pumping is different from reality.

Decreasing pump speed is analogous to using a smaller impeller size in the volute of the pump, and the result is that motor power is reduced by the cube of the speed reduction, closely following the affinity laws. (Variable-speed pump curves are shown in Chapter 43.) Design flow conditions generally occur for minimal hours of operation per year; as such, the energy savings potential for a variable-speed pump is great. In the general control valve application, a reasonably selected equal percentage valve with a 50% valve authority should reduce flow by about 30% when the valve is positioned from 100% open (design flow, minimal hours per year) to 90% open. The 10% change in stroke should reduce pump operating power about 70%. Hydronic systems should operate most of the time at flow rates well below design. The coil characteristic suggests that, in sensible applications, a small percentage of flow yields an exceptionally high degree of coil design heat transfer, adequate for most of the year's operation.

Depending on system design, direct digital control may also allow more advanced control strategies. There are various reset control strategies (e.g., cascade control) to optimize pump speed and flow performance. Many of these monitor valve position and drive the pump to a level that keeps one valve open while maintaining set point for comfort conditions. Physical operational requirements of the components must be taken into account. There are some concerns over operating pumps and their motor drives at speeds less than 30% of design, particularly about maintaining proper lubrication of the pump mechanical seals and motor bearings. From a practical perspective, 30% speed is a scant 3% of design power, so it may be unnecessary to reduce speed any further. Consult manufacturers for information on device limits, to maintain the system in good operating condition.

Exceptional energy reduction potential and advances in variable-frequency drive technology that have reduced drive costs have made

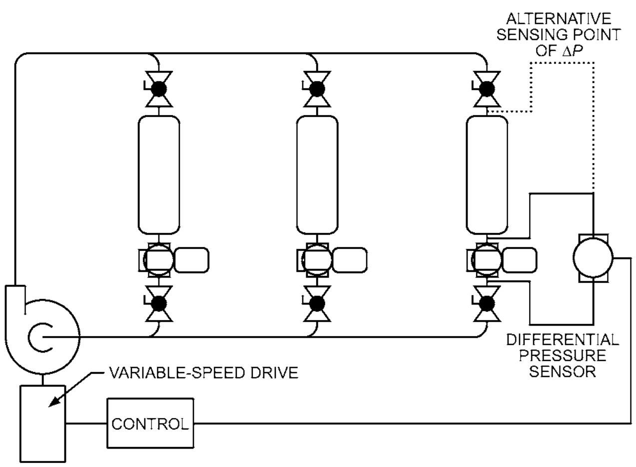

Fig. 11 Example of Variable-Speed Pump System Schematic

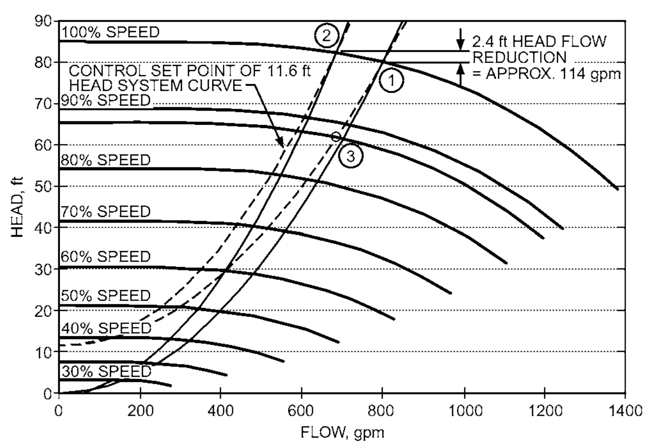

Fig. 12 Example of Variable-Speed Pump and System Curves

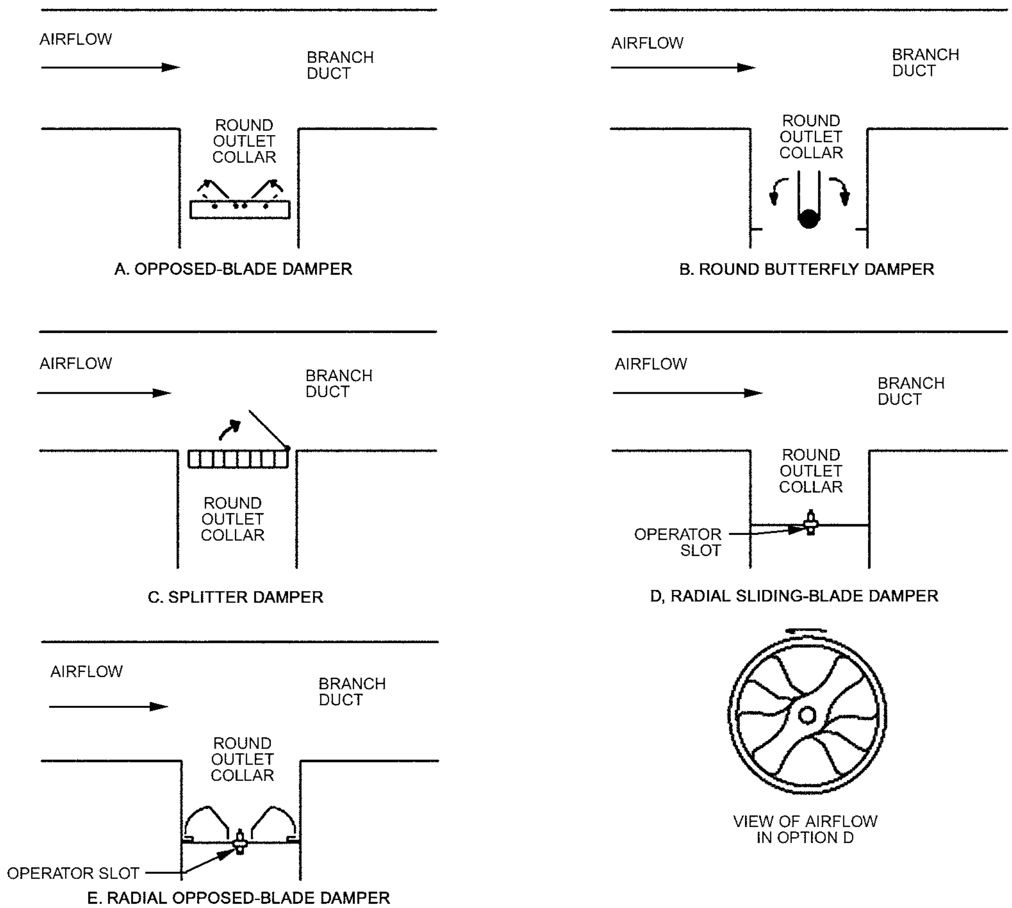

Fig. 4 Accessory Controls for Ceiling Diffusers

tem pressure requirement, thereby limiting downstream airflow and increasing the sound level. An extractor mounted directly to the grille (Figure 3F) is sometimes called a **scoop**.

Adjustable vanes, as shown in Figure 3E, are individually adjustable. Typically, two sets of vanes are used. One set equalizes flow across the collar, and the other set turns the air. The vanes should not be adjusted to act as a damper, because balancing requires removing the grille to gain access and then reinstalling it to measure airflow.

Equalizing or **flow-straightening devices** or **grids** allow adjustment of the airstream to obtain more uniform flow to the diffuser.

Quadrant blank-off flat plates restrict airflow in one or more quadrant(s) of the diffuser inlet.

Other balancing devices are available. Consult manufacturers' literature as a source of information for other air-balancing devices.

3.6 TERMINAL UNITS

In air distribution systems, special control and acoustical equipment is frequently required to introduce conditioned air into a space properly. Airflow controls for these systems consist principally of terminal units (historically called **VAV boxes**), with which the airflow can be varied by modulating valves, fan controls, or both. Terminal units such as those listed here are supplied with or without cooling, fans, heat, or reheat, and having either constant or variable primary airflow rate. Terminal units typically control the flow of conditioned air supplied through a duct from an air handler and deliver it to the occupied space, but sometimes, such as in the case of underfloor units located in a cooled supply air plenum, may take the air directly from the plenum with no primary duct. They may use a ducted or nonducted return plenum or room air induction to affect space temperature control while maintaining a constant or variable discharge airflow rate and/or temperature. Terminal units may include sound reduction devices, heaters, reheaters, fans, diffusers, or cooling coils.

This section discusses control equipment for forced air-conditioning systems. Chapter 49 of the 2019 *ASHRAE Handbook—HVAC Applications* includes information on sound control in air-conditioning systems and sound rating for air outlets.

Terminal units are factory-made assemblies for air distribution. A terminal unit manually or automatically performs one or more of the following functions: (1) controls air velocity, airflow rate, pressure, or temperature; (2) mixes primary air from the duct system with air from the treated space or from a secondary duct system; and (3) heats or cools the air. To achieve these functions, terminal unit assemblies are made from an appropriate selection of the following components: casing, mixing section, manual or automatic air control device, heat exchanger, induction section (with or without fan), sound reduction devices, and flow controller.

Terminal units are typically classified as constant- or variable-volume devices. They are further categorized as either pressure-dependent, where airflow through the assembly varies in response to changes in system pressure, or pressure-independent (pressure-compensating), where airflow through the device does not vary in response to changes in system pressure.

the coil is adjusted by varying output flow through the coil. In some limited applications, such as a cooling tower control, a diverting or bypass valve must be used in place of a mixing valve. In most cases, a mixing valve can perform the same function as a diverting or bypass valve if the companion actuator has a very high spring rate. Otherwise, water hammer or noise may occur when operating near the seat.

Special-Purpose Valves

Special-purpose valve bodies may be used on occasion. One type of four-way valve is used to allow separate circulation in the boiler loop and a heated zone. Another type of four-way valve body is used as a changeover refrigeration valve in heat pump systems to reverse the evaporator to a condenser function. Six-way valves are commonly used with four-pipe chilled-beam systems having two coil connections (supply and return), which limits output to the chilled beam to either heating, cooling, or neither.

Float valves are used to supply water to a tank or reservoir or serve as a boiler feed valve to maintain an operating water level at the float level location (Figure 13).

Ball Valves

Ball valves coupled with electronic direct-coupled actuators are common in HVAC control applications because it is easy to match the movement of the 90° actuator travel to the quarter-turn movement of the ball valve. For modulating flow applications, a reduced port or characterizing disc should be used on the ball valve to reduce the overall flow capacity C_v of the valve and modify the flow characteristic (e.g., equal percentage) to prevent overflow or uneven flow. The packing system of a characterized ball valve is designed to accommodate the modulating control action inherent in HVAC.

Butterfly Valves

Butterfly valves are used in two-position and modulating applications. Advantages of butterfly valves are their ability to provide a bubbletight close off against higher pressures, their lighter weight, and compact size.

Butterfly valves are manufactured as two-way valves. In some applications where there is a need for a three-way valve application, two butterfly valves are connected to a piping tee and cross-linked to operate as either three-way mixing or three-way bypass valves (Figure 14).

Butterfly valves have different flow characteristics from standard seat and disk-type globe valves, so they may be used only where their flow characteristics suffice. When sizing a butterfly valve for a modulating application, use the C_v rating at 60° or 70° rotation; for two-position (on/off) applications, use the 90° C_v rating. It is also important to follow the manufacturer's guideline for the butterfly valve rating for maximum fluid velocity, because exceeding this figure can shorten the valve's life expectancy.

Pressure-Independent Control Valves

Pressure-independent control valves (PICVs; Figure 15) are control valves coupled with an internal method of differential pressure regulation to eliminate variances of flow caused by differential pressure fluctuations across the valve assembly. This can be accomplished by using a dynamic pressure regulator or integrated software to monitor the flow in relation to the designed output of the valve similar to a pressure-independent variable-air-volume (VAV) box.

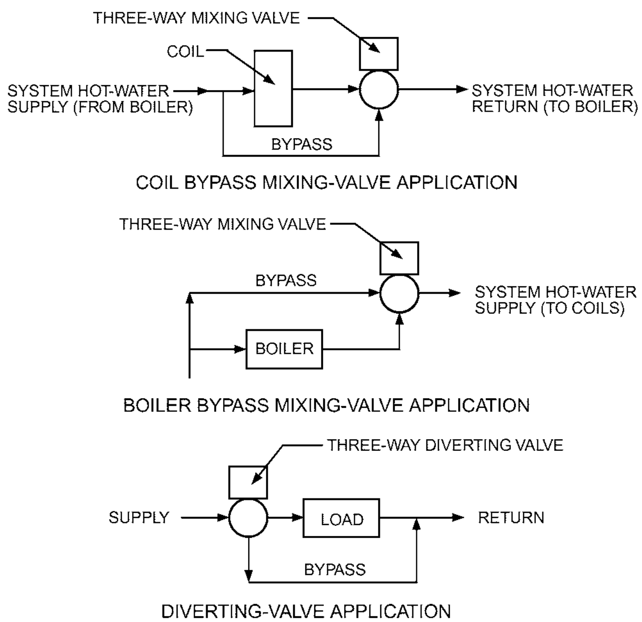

Fig. 12 Typical Three-Way Control Applications

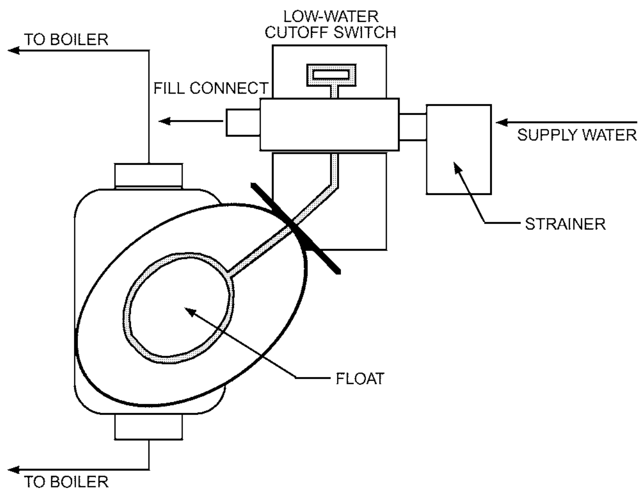

Fig. 13 Float Valve and Cutoff Steam Boiler Application

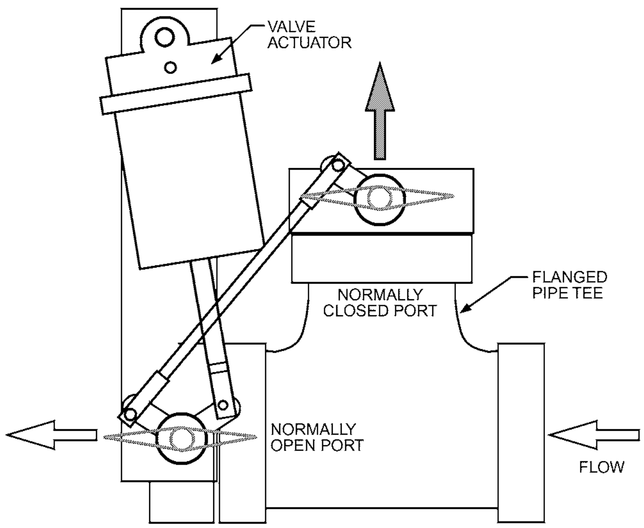

Fig. 14 Butterfly Valves, Diverting Tee Application

In practical application, a PICV replaces both a control valve and a balancing device.

The flow of standard pressure-dependent control valves (globe valves, ball valves, etc.) depends on the pressure differential across the valve. As this pressure rises across a standard valve with a fixed stem position, the flow increases. Likewise, as differential pressure across the valve drops, the flow decreases. This change in pressure across the control valve is caused by factors such as the modulation of other control valves in the hydronic loop. Additionally, varying differential pressure across a control valve results in over- and underflow caused by rapid changes in flow with small changes in valve positioning (see the section on Control Valve Flow Characteristics).

Advantages of PICVs include stable flow when the stem position is fixed, regardless of pressure fluctuations, and the elimination of a hydronic balancing device (e.g. balancing valve, flow-limiting device) where the PICV is mounted. Sizing a PICV is straightforward because it is no longer necessary to calculate a C_v for the valve. PICV selection is always based on the full-open valve flow required. Ultimately, PICVs operate as if they have a perfect valve authority, even though having a perfect valve authority is mathematically impossible.

PICVs are rated for a differential pressure range (e.g., 5 to 50 psi) where the PICV is pressure independent. If the pressure falls outside this range, the PICV will become a pressure-dependent valve, susceptible to flow variations from pressure changes. This differential pressure range is an operating range unrelated to the close-off pressure of the valve itself.

PICVs can be field adjustable for their maximum flow set point by either adjusting actuator end stops or programming, thereby eliminating the need to remove and replace components. The only limitation to adjusting the flow is the capacity of the valve body itself.

Flow-Limiting Valves

Flow-limiting valves (FLVs) are control valves coupled with an integrated flow-limiting device that prevents the flow from exceeding a predetermined limit. Flow is limited by an internal cartridge assembly. The FLV will not allow the flow to exceed the flow limit when operating within the predefined differential pressure operating range (e.g., 1 to 14 psi, 2 to 32 psi). However, unlike a pressure-independent control valve that eliminates flow variations (i.e., the over- and underflow caused by pressure across the entire opening range of the control valve), FLVs essentially do nothing when flow is below design when the valve is throttling flow, and do nothing to limit differential pressure across the control valve (Peterson 2017).

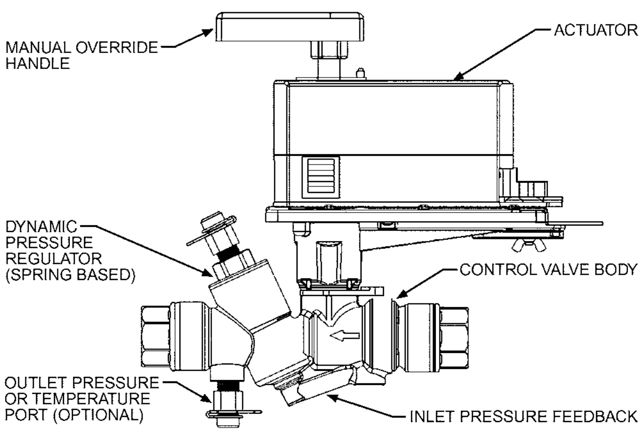

Fig. 15 Pressure-Independent Control Valve

FLVs are for two-position (on/off) applications where the design calls for full flow or no flow, with no modulation in between.

The flow limit of FLVs can be field adjusted by (1) replacing their internal cartridge or (2) external adjustment, with the only limitation being the flow capacity of the valve body.

Control Valve Flow Characteristics

Based on the characteristics of the valve, three distinct flow conditions are possible (Figure 16):

- **Quick Opening.** When started from the closed position, a quick-opening valve allows a considerable amount of flow through a small opening of the valve. As the valve moves toward the full open position, the rate at which the flow is increased is reduced in a nonlinear fashion. This characteristic is used in two-position or on/off applications, because it is very difficult to position the stem accurately enough for low and medium flows in modulating applications.
- **Linear.** Linear valves produce equal flow increments per equal increments of the valve moving towards the full open position. This characteristic is commonly used on steam coil terminals, bypass applications, and sometimes in the bypass port of three-way valves.
- **Equal Percentage.** This type of valve produces an exponential flow increase as the valve moves from the closed to the open position. For equal increments of the valve opening, the flow increases by an equal percentage. For example, in Figure 16, if the valve modulates from 50 to 70% of full stroke, the percentage of full flow changes from 10 to 25%, an increase of 150%. Then, if the valve modulates from 80 to 100% of full stroke, the percentage of full flow changes from 40 to 100%, again, an increase of 150%. This characteristic is recommended for control of hot- and chilled-water coils to linearize the output of the coil-valve combination, which makes control tuning easier and typically results in more stable systems.

The designer combines the valve flow characteristic with the coil performance curve (heating or cooling) to ensure the proper heat transfer characteristic. (Figure 17). The coil output (Figure 17A) of a heating or cooling coil increases rapidly and begins to slow as it

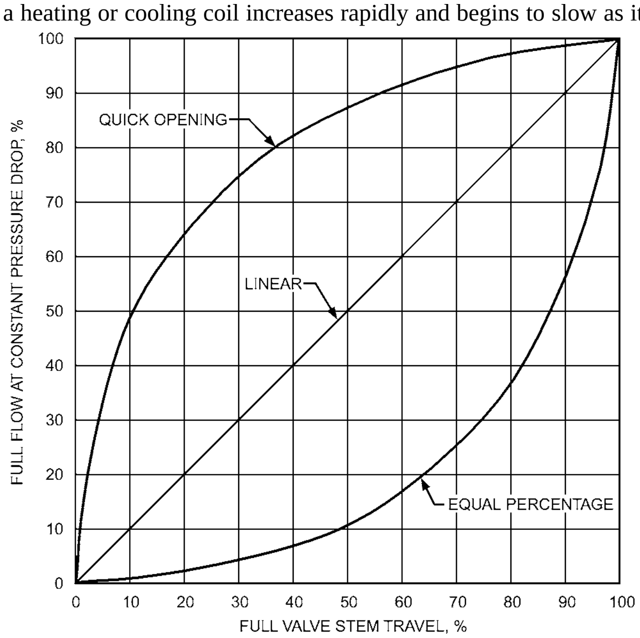

Fig. 16 Control Valve Flow Characteristics